

Network Access Device:

- o NAD stands for Network Access Device.
- o Network Access Device (NAD) is the Access Layer Device.
- o NAD is basically, a device through which Endpoint is connected.
- o Network Access Device (NAD) is AAA Client device.
- o Any device that is going to send RADIUS authentication requests to ISE.
- o NAD responsible for sending EAP authentication request to authentication server.
- o NAD is Authenticator for 802.1X from a supplicant to the authentication server.
- o NAD can be the device such as Switch, Router that support NAC Enforcement policies.
- o Common NAD types Wired Ethernet Switches, Wireless LAN Controllers & ASA.



Wireless LAN Controller

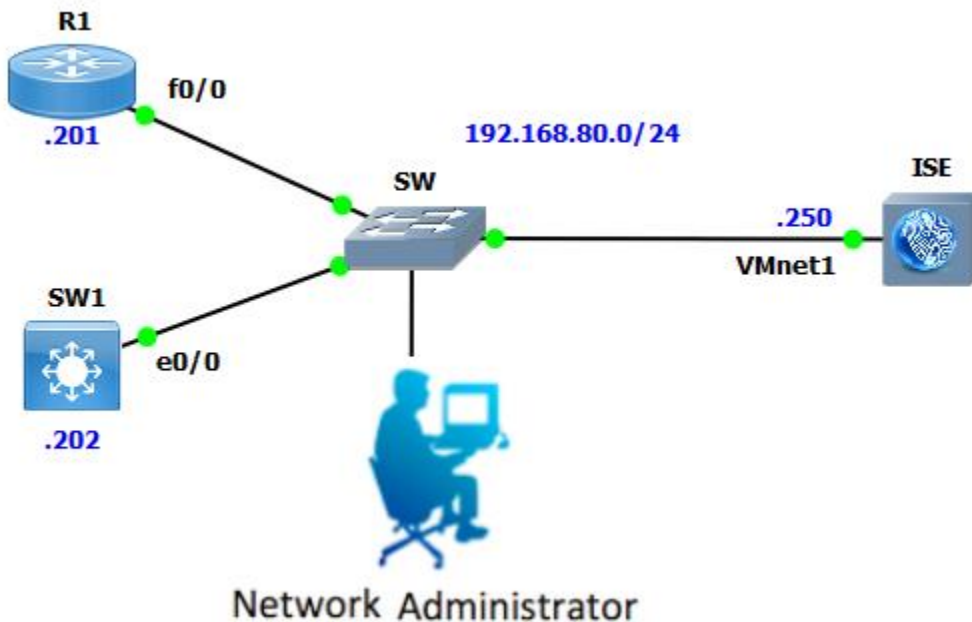
Switch

Router

The screenshot shows the Cisco Identity Services Engine (ISE) Administration interface. The navigation pane on the left includes 'Administration' (1), 'Network Resources' (2), and 'Network Devices' (3). Under 'Network Devices', 'Network devices' (4) is selected. The main pane shows the 'Network Devices' table (5) with columns: Name, IP/Mask, Profile Name, and Location. A device named 'SW1' with IP '192.168.80.1...' and Profile Name 'Cisco' is listed (6). The 'Add' button (5) is highlighted in the toolbar above the table.

Name	IP/Mask	Profile Name	Location
SW1	192.168.80.1...	Cisco	All L...

Network Diagram:



Configure Router for Authentication and Authorization:

Enable AAA new-model. Define TACACS server ISE. Configure login, enable authentications, and then use the exec and command authorizations. Assign method lists to line vty.

R1 AAA Configuration
R1(config)#aaa new-model
R1(config)#tacacs-server host 192.168.80.250 key test123
R1(config)#aaa authentication login default group tacacs+ local
R1(config)#aaa authentication enable default group tacacs+ enable
R1(config)#aaa authorization exec default group tacacs+ local
R1(config)#aaa authorization commands 0 default group tacacs+ local
R1(config)#aaa authorization commands 1 default group tacacs+ local
R1(config)#aaa authorization commands 15 default group tacacs+ local
R1(config)#aaa authorization config-commands
R1(config)#line vty 0 4
R1(config-line)#authorization commands 0 default
R1(config-line)#authorization commands 1 default
R1(config-line)#authorization commands 15 default
R1(config-line)#authorization exec default
R1(config-line)#login authentication default

Configure Switch for Authentication and Authorization:

Enable AAA new-model. Define TACACS server ISE. Configure login, enable authentications, and then use the exec and command authorizations. Assign method lists to line vty.

SW1 AAA Configuration
SW1(config)#aaa new-model
SW1(config)#tacacs server ISE SW1(config-server-tacacs)#address ipv4 192.168.80.250 SW1(config-server-tacacs)#key test123
SW1(config)#aaa authentication login default group tacacs+ local SW1(config)#aaa authentication enable default group tacacs+ enable SW1(config)#aaa authorization exec default group tacacs+ local SW1(config)#aaa authorization commands 0 default group tacacs+ local SW1(config)#aaa authorization commands 1 default group tacacs+ local SW1(config)#aaa authorization commands 15 default group tacacs+ local SW1(config)#aaa authorization config-commands
SW1(config)#line vty 0 4 SW1(config-line)#authorization commands 0 default SW1(config-line)#authorization commands 1 default SW1(config-line)#authorization commands 15 default SW1(config-line)#authorization exec default SW1(config-line)#login authentication default

Add Local Groups & Users:

Navigate to **Administration > Identity Management > Groups > User Identity Groups**. Click **Add**. Provide Name, Description and click “Save” to save setting.

The screenshot displays the Cisco Identity Services Engine (ISE) Administration interface. The top navigation bar includes 'Home', 'Operations', 'Policy', 'Guest Access', and 'Administration' (highlighted with a red box). Below this, the 'Identity Management' section is expanded, showing 'Groups' (highlighted with a red box) and 'User Identity Groups' (highlighted with a red box). The main content area shows the 'User Identity Groups > AAA-Group' configuration page. The 'Identity Group' section has a '* Name' field with 'Admin-Group' (highlighted with a red box) and a 'Description' field with 'This is Admin Group'. At the bottom, there are 'Save' and 'Reset' buttons, with 'Save' highlighted by a red box. The 'Member Users' section is visible below the buttons.

Navigate to **Administration > Identity Management > Groups > User Identity Groups**. Click **Add**. Provide Name, Description and click **“Save”** to save setting.

The screenshot shows the Cisco ISE Administration console. The top navigation bar includes 'Administration' (1). The left sidebar shows 'Identity Management' (2) > 'Groups' (3) > 'User Identity Groups' (4). The main content area is titled 'User Identity Groups > New User Identity Group'. It contains a form with the following fields:

- '* Name' (5) with the value 'Support-Group'.
- 'Description' (7) with the value 'This is Support Group only Show Commands' (6).
- 'Submit' and 'Cancel' buttons.

Navigate to **Administration > Identity Management > Identities > Users**. Click **Add**. Provide Name, Passwords and User Groups select Admin-Group and click **“Save”** to save the setting.

The screenshot shows the Cisco ISE Administration console. The top navigation bar includes 'Administration' (1). The left sidebar shows 'Identity Management' (2) > 'Identities' (3) > 'Users' (4). The main content area is titled 'Network Access Users List > admin1'. It contains a form with the following sections:

- '* Name' (5) with the value 'admin1'.
- 'Status' set to 'Enabled'.
- 'Email' field.
- 'Passwords' (6) section with:
 - '* Login Password' field.
 - 'Re-Enter Password' field.
 - 'Enable Password' field.
- 'User Information' section.
- 'Account Options' section.
- 'User Groups' (7) section with a dropdown menu showing 'Admin-Group' (8).
- 'Save' and 'Reset' buttons.

Navigate to **Administration > Identity Management > Identities > Users**. Click **Add**. Provide Name, Passwords and User Groups select Support-Group and click **“Save”** to save the setting.

The screenshot displays the Cisco Identity Services Engine (ISE) Administration console. The navigation path is highlighted with red boxes and numbers: 1. Administration (top bar), 2. Identity Management (left sidebar), 3. Identities (left sidebar), 4. Users (left sidebar), 5. Network Access User (main header), 6. Passwords (section header), 7. User Groups (section header), and 8. Submit (button). The form fields are as follows:

- Network Access User** (Section 5)
 - * Name: support1
 - Status: ☒ Enabled
 - Email:
- Passwords** (Section 6)
 - * Login Password: (masked) Re-Enter Password: (masked)
 - Enable Password: (checkbox) (masked)
- User Information** (Section)
- Account Options** (Section)
- User Groups** (Section 7)
 - Support-Group (selected)
- Buttons** (Section 8): Submit, Cancel

Add Network Device:

Navigate to **Administration > Network Resources > Network Devices**. Click **Add**. Provide Name, IP Address, select **TACACS+ Authentication Settings** checkbox and provide Shared Secret key.

The screenshot displays the Cisco Identity Services Engine (ISE) web interface for adding a new network device. The navigation path is **Administration > Network Resources > Network Devices**. The form includes the following fields and options:

- Name:** R1
- Description:** R1
- * IP Address:** 192.168.80.201 / 32
- TACACS+ Authentication Settings:** ☒
- Shared Secret:** [Masked with dots] [Show]
- Enable Single Connect Mode:** ☐
- Legacy Cisco Device:** ☒
- TACACS+ Draft Compliance Single Connect Support:** ☐
- SNMP Settings:** ☐
- Advanced TrustSec Settings:** ☐
- Buttons:** Save, Reset

Navigate to **Administration > Network Resources > Network Devices**. Click **Add**. Provide Name, IP Address, select **TACACS+ Authentication Settings** checkbox and provide Shared Secret key.

Identity Services Engine Home Operations Policy Guest Access Administration

System Identity Management Network Resources Device Portal Management pxGrid Services Feed S

Network Devices Network Device Groups Network Device Profiles External RADIUS Servers RADIUS Server Seq

Network devices

Default Device

Network Devices List > R1

Network Devices

* Name S1

Description S1

* IP Address: 192.168.80.202 / 32

☒ TACACS+ Authentication Settings

Shared Secret

Show

Enable Single Connect Mode ☐

Legacy Cisco Device

TACACS+ Draft Compliance Single Connect Support

☐ SNMP Settings

☐ Advanced TrustSec Settings

Save Reset

Enable Device Admin Service:

Navigate to **Administration > System > Deployment**. Select required Node. Select **Enable Device Admin Service** checkbox and click **“Save”** to save the setting:

Configuring TACACS Command Sets:

First **PermitAllCommands** for the user admin, which allow all commands on the device. Second **PermitShowCommands** for user support, which will allow only show commands.

1. Navigate to **Work Centers > Device Administration > Policy Results > TACACS Command Sets**. Click **Add**. Provide the Name **PermitAllCommands**, select **Permit any command** checkbox that is not listed below and click **Submit**.

2. Navigate to **Work Centers > Device Administration > Policy Results > TACACS Command Sets**. Click **Add**. Provide the Name **PermitShowCommands**, click **Add** and permit **show** and **exit** commands. By default if Arguments is left blank, all arguments are be included. Click **Submit**.

Identity Services Engine Home Operations Policy Guest Access Administration Work Centers

TrustSec Device Administration

Overview Identities User Identity Groups Network Resources Network Device Groups Policy Conditions Policy Results

TACACS Command Sets

TACACS Profiles

TACACS Command Sets > PermitShowCommands

Command Set

Name * PermitShowCommands

Description

Permit any command that is not listed below

+ Add Trash Edit Move Up Move Down

Grant	Command	Arguments
☐	PERMIT	exit
☐	PERMIT	show

Configuring TACACS Profile:

Navigate to **Work Centers > Device Administration > Policy Results > TACACS Profiles**. Click **Add**. Provide Name **ShellProfile**, select **Default Privilege** checkbox and enter the value of 15. Click **Submit**.

Identity Services Engine Home Operations Policy Guest Access Administration Work Centers

TrustSec Device Administration

Overview Identities User Identity Groups Network Resources Network Device Groups Policy Conditions Policy Results

TACACS Command Sets

TACACS Profiles

TACACS Profiles > ShellProfile

TACACS Profile

Name * ShellProfile

Description

Task Attribute View Raw View

Common Tasks

☒ Default Privilege 15 Select 0 to 15)

Configuring TACACS Authorization Policy:

Navigate to **Work Centers > Device Administration > Policy Sets > Default > Authorization Policy > Edit > Insert New Rule Above**.

Two authorization rules are configured, first rule assigns TACACS profile **ShellProfile** and command Set **PermitAllCommands** based on Network **Admin-Group** membership. Second rule assigns TACACS profile **ShellProfile** and command Set **PermitShowCommands** based on Network **Support-Group** membership.

The screenshot shows the Cisco ISE Policy Sets configuration page. The 'Default' policy set is selected, and the 'Authorization Policy' tab is active. Two rules are configured:

Status	Rule Name	Conditions (identity groups and other conditions)	Command Sets	Shell Profiles
✓	PermitAllCommands	if Admin-Group	then PermitAllCommands AND ShellProfile	
✓	PermitShowCommands	if Support-Group	then PermitShowCommands AND ShellProfile	
✓	Tacacs_Default	if no matches, then	DenyAllCommands	

Verify:

Navigate to **Operations > TACACS Livelog**.

The screenshot shows the Cisco ISE TACACS Livelog page. The table displays log entries for TACACS authentication and authorization. The first entry shows a failed authorization attempt for 'support1' with a red 'X' in the status column. The second entry shows a successful authorization attempt for 'support1' with a green checkmark in the status column.

Logged Time	Status	Details	Username	Type	Authentication Policy	Authorization Policy	ISE Node
2019-06-07 08:54:40.406	✗		support1	Authorization		Tacacs_Default >> PermitShowC...	ISE1
2019-06-07 08:54:38.46	✓		support1	Authorization		Tacacs_Default >> PermitShowC...	ISE1
2019-06-07 08:54:38.417	✓		support1	Authentication	Tacacs_Default >> Default >> Default		ISE1
2019-06-07 08:54:29.839	✓		admin1	Authorization		Tacacs_Default >> PermitAllCom...	ISE1
2019-06-07 08:54:23.9	✓		admin1	Authorization		Tacacs_Default >> PermitAllCom...	ISE1
2019-06-07 08:54:21.138	✓		admin1	Authorization		Tacacs_Default >> PermitAllCom...	ISE1
2019-06-07 08:54:18.395	✓		admin1	Authorization		Tacacs_Default >> PermitAllCom...	ISE1
2019-06-07 08:54:18.296	✓		admin1	Authentication	Tacacs_Default >> Default >> Default		ISE1

Cisco IOS Router Verification.

```
R1#telnet 192.168.80.202
Trying 192.168.80.202 ... Open
Username:support1
Password:
```

```
SW1#config t
Command authorization failed.
```